

Linea Audit Summary

Prepared by Cyfrin

Version 2.2

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Linea on Linea. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.2 dated January 6, 2025, was delivered to Linea, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 11 days, the Cyfrin team conducted an audit on the Linea code provided by Linea. The findings consist of 1 Medium & 1 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Linea
Repository	linea-monorepo
Commit	5fad7f793cc1...
Audit Timeline	Oct 21st - Nov 4th, 2024
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	1	1	0
Low Risk	1	1	0
Informational	5	5	0
Gas Optimizations	5	5	0
Total	12	12	0

6 Audit Scope

The audit scope was limited to Solidity smart contracts primarily covering the messaging service, token bridge and rollup, aiming to audit the following new developments:

- changes to events emitted during blob submission and finalization to simplify L2 state reconstruction and remove obsolete elements
- more granular role-based access control
- changes to TokenBridge storage
- a new fallback finalization feature
- removal of the previous “training wheel” finalization feature
- increase of Solidity version to 0.8.26
- optimized hashing during message sending and claiming
- assorted minor improvements
- additional commits e62791d, a83412e, 94fc031, 448a4f4, f0bc3fa, 63357ef, 7918d8c, c9e67e7
- PRs 298, 299, 300, 301, 302, 303, 304, 305, 306, 307 from OpenZeppelin’s audit incorporated into commit adb097a

The final commit hash with all fixes included was a83412e which was used to tag release contract-audit-2025-01-06.

All code relating to zk provers and other code outside the smart contract layer was out of scope. The following contracts were included in the scope for this audit:

```
contracts/LineaRollup.sol
contracts/ZkEvmV2.sol
contracts/interfaces/IGenericErrors.sol
contracts/interfaces/IMessageService.sol
contracts/interfaces/IPauseManager.sol
contracts/interfaces/IPermissionsManager.sol
contracts/interfaces/IRateLimiter.sol
contracts/interfaces/l1/IL1MessageManager.sol
contracts/interfaces/l1/IL1MessageManagerV1.sol
contracts/interfaces/l1/IL1MessageService.sol
contracts/interfaces/l1/ILineaRollup.sol
contracts/interfaces/l1/IPlonkVerifier.sol
contracts/interfaces/l1/IZkEvmV2.sol
contracts/interfaces/l2/IL2MessageManager.sol
contracts/interfaces/l2/IL2MessageManagerV1.sol
contracts/lib/L2MessageServicePauseManager.sol
contracts/lib/LineaRollupPauseManager.sol
contracts/lib/PermissionsManager.sol
contracts/lib/TokenBridgePauseManager.sol
contracts/lib/PauseManager.sol
contracts/lib/Utils.sol
contracts/messageService/MessageServiceBase.sol
contracts/messageService/l1/L1MessageManager.sol
contracts/messageService/l1/L1MessageService.sol
contracts/messageService/l1/TransientStorageReentrancyGuardUpgradeable.sol
contracts/messageService/l1/v1/L1MessageManagerV1.sol
contracts/messageService/l1/v1/L1MessageServiceV1.sol
contracts/messageService/l2/L2MessageManager.sol
contracts/messageService/l2/L2MessageService.sol
contracts/messageService/l2/v1/L2MessageManagerV1.sol
contracts/messageService/l2/v1/L2MessageServiceV1.sol
contracts/messageService/lib/MessageHashing.sol
contracts/messageService/lib/RateLimiter.sol
contracts/messageService/lib/SparseMerkleTreeVerifier.sol
contracts/messageService/lib/TransientStorageHelpers.sol
contracts/tokenBridge/TokenBridge.sol
contracts/tokenBridge/interfaces/ITokenBridge.sol
contracts/tokenBridge/lib/StorageFiller39.sol
```

7 Deployment Bytecode Verification

At the conclusion of the audit, after all mitigations were concluded, Cyfrin was asked to verify that at commit df30415 the locally compiled bytecode matched the expected bytecode listed in the repository and the deployed bytecode on mainnet. The following reproducible process was used.

First, prepare the repository:

```
gh repo clone Consensys/linea-monorepo
cd linea-monorepo
git checkout df304158985a85b9f80980475fbd74d7d3802034
make pnpm-install
cd contracts
make test
```

Second, ensure that no in-scope contracts changed between the bytecode verification commit df30415 and the last commit reviewed within the audit scope, adb097a:

```
cd ..
git diff adb097aff4d7d32da843b16c9e1c1b21eecbf955 | grep -i "contracts/contracts/"
```

Then, for each contract below, extract `deployedBytecode` from the Hardhat build artifact (`contracts/build/contracts/.../<Contract>.json`) into a `<Contract>.local.txt` file and the deployed bytecode from the block explorer into a `<Contract>.mainnet.txt` file in `contracts/deployments/bytecode/2024-12-03/`, copy the header information from the expected bytecode file `<Contract>.txt` into both, and `diff` the expected file against each of them to verify everything matches:

- `LineaRollup`: 0x07ddce60658a61dc1732cacf2220fce4a01c49b0
- `L2MessageService`: 0x05d43713b7e333d2d54be65ce3b5f3698ab960fd
- L2 `TokenBridge`: 0xD90ed3D4f9d11262d3D346a4369058d5B3777137
- L1 `TokenBridge`: 0x2B6A2F8880220a66DfB9059FCB76F7dB54104a34

For the L1 `TokenBridge`, first edit `contracts/hardhat.config.ts` to comment out `evmVersion` for solc `0.8.19` and compile again using `npx hardhat compile`:

```
{
  version: "0.8.19",
  settings: {
    viaIR: useViaIR,
    optimizer: {
      enabled: true,
      runs: 10_000,
    },
    //evmVersion: "london",
  },
},
```

8 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

9 Protocol Summary

Linea is a Type 2 zkEVM L2 rollup with full EVM equivalence that aims to provide the “Ethereum experience” at scale.

10 Attestation

Cyfrin attests that it performed a security review of the Linea code provided by Linea.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-01-06-cyfrin-linea-v2.2.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Linea has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=2bdc94162d17ae59cb9152434cac71357ebe6120deb3b9baf159063eab05bbd5
F=2025-01-06-cyfrin-linea-v2.2-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.